APPENDIX C

Levering and Unlevering the Cost of Equity

This appendix derives various formulas that can be used to compute unlevered beta and the unlevered cost of equity under different assumptions. Unlevered betas are required to estimate an industry beta, as detailed in Chapter 15. We prefer using an industry beta rather than a company beta to determine the cost of capital because company betas cannot be estimated accurately. As discussed in Chapter 10, the unlevered cost of equity is used to discount free cash flow to compute adjusted present value. For companies with substantial postretirement obligations, the appendix concludes by incorporating pensions and other postretirement benefits into the unlevering process.

UNLEVERED COST OF EQUITY

Franco Modigliani and Merton Miller postulated that the market value of a company's economic assets, such as operating assets (V_u) and tax shields (V_{txa}), should equal the market value of its financial claims, such as debt (D) and equity (E):

$$V_u + V_{txa} = \text{Enterprise Value} = D + E \qquad \text{(C.1)}$$

A second result of Modigliani and Miller's work is that the total risk of the company's economic assets, operating and financial, must equal the total risk of the financial claims against those assets:

$$\frac{V_u}{V_u + V_{txa}}(k_u) + \frac{V_{txa}}{V_u + V_{txa}}(k_{txa}) = \frac{D}{D+E}(k_d) + \frac{E}{D+E}(k_e) \qquad \text{(C.2)}$$

where k_u = unlevered cost of equity
k_{txa} = cost of capital for the company's interest tax shields
k_d = cost of debt
k_e = cost of equity

The four terms in this equation represent the proportional risk of operating assets, tax assets, debt, and equity, respectively.

Since the cost of operating assets (k_u) is unobservable, it is necessary to solve for it using the equation's other inputs. The required return on tax shields (k_{txa}) also is unobservable. With two unknowns and only one equation, it is therefore necessary to impose additional restrictions to solve for k_u. If debt is a constant proportion of enterprise value (i.e., debt grows as the business grows), k_{txa} equals k_u. Imposing this restriction leads to the following equation:

$$\frac{V_u}{V_u+V_{txa}}(k_u)+\frac{V_{txa}}{V_u+V_{txa}}(k_u)=\frac{D}{D+E}(k_d)+\frac{E}{D+E}(k_e)$$

Combining terms on the left side generates an equation for the unlevered cost of equity when debt is a constant proportion of enterprise value:

$$k_u=\frac{D}{D+E}(k_d)+\frac{E}{D+E}(k_e) \tag{C.3}$$

Since most companies manage their debt-to-value ratio to stay within a particular range, we believe this formula and its resulting derivations are the most appropriate for standard valuation.

Unlevered Cost of Equity When k_{txa} Equals k_d

Some financial analysts set the required return on interest tax shields equal to the cost of debt. In this case, Equation C.2 can be expressed as follows:

$$\frac{V_u}{V_u+V_{txa}}(k_u)+\frac{V_{txa}}{V_u+V_{txa}}(k_d)=\frac{D}{D+E}(k_d)+\frac{E}{D+E}(k_e)$$

To solve for k_u, multiply both sides by enterprise value:

$$V_u(k_u)+V_{txa}(k_d)=D(k_d)+E(k_e)$$

and move $V_{txa}(k_d)$ to the right side of the equation:

$$V_u(k_u)=(D-V_{txa})k_d+E(k_e)$$

EXHIBIT C.1 **Unlevered Cost of Equity**

	Dollar level of debt fluctuates	Dollar level of debt is constant
Tax shields have same risk as operating assets $k_{txa} = k_u$	$k_u = \frac{D}{D+E} k_d + \frac{E}{D+E} k_e$	$k_u = \frac{D}{D+E} k_d + \frac{E}{D+E} k_e$
Tax shields have same risk as debt $k_{txa} = k_d$	$k_u = \frac{D - V_{txa}}{D - V_{txa} + E} k_d + \frac{E}{D - V_{txa} + E} k_e$	$k_u = \frac{D(1-T_m)}{D(1-T_m)+E} k_d + \frac{E}{D(1-T_m)+E} k_e$

Note: k_e = cost of equity
k_d = cost of debt
k_u = unlevered cost of equity
k_{txa} = cost of capital for tax shields
T_m = marginal tax rate
D = debt
E = equity
V_{txa} = present value of tax shields

To eliminate V_u from the left side of the equation, rearrange Equation C.1 to $V_u = D - V_{txa} + E$, and divide both sides by this value:

$$k_u = \frac{D - V_{txa}}{D - V_{txa} + E}(k_d) + \frac{E}{D - V_{txa} + E}(k_e) \tag{C.4}$$

Equation C.4 mirrors Equation C.2 closely. It differs from Equation C.2 only in that the market value of debt is reduced by the present value of expected tax shields.

Unlevered Cost of Equity When Debt Is Constant

Exhibit C.1 summarizes three methods to estimate the unlevered cost of equity. The two formulas in the top row assume that the risk associated with interest tax shields (k_{txa}) equals the risk of operations (k_u). When this is true, whether debt is constant or expected to change, the formula remains the same.

The bottom-row formulas assume that the risk of interest tax shields equals the risk of debt. On the left, future debt can take on any value. On the right, an additional restriction is imposed that debt remains constant—in absolute terms, not as a percentage of enterprise value. In this case, the annual interest payment equals $D(k_d)$, and the annual tax shield equals $D(k_d)(T_m)$. Since tax shields are constant, they can be valued using a constant perpetuity:

$$\text{PV}(\text{Tax Shields}) = \frac{D(k_d)(T_m)}{k_d} = D(T_m)$$

Consequently, V_{txa} in the formula in the bottom left corner is replaced with $D(T_m)$. The equation is simplified by converting $D - D(T_m)$ into $D(1 - T_m)$. The resulting equation is presented in the bottom right corner.

LEVERED COST OF EQUITY

In certain situations, you will have already estimated the unlevered cost of equity and need to relever the cost of equity to a new target structure. In this case, use Equation C.2 to solve for the levered cost of equity, k_e:

$$\frac{V_u}{V_u + V_{txa}}(k_u) + \frac{V_{txa}}{V_u + V_{txa}}(k_{txa}) = \frac{D}{D+E}(k_d) + \frac{E}{D+E}(k_e)$$

Multiply both sides by enterprise value:

$$V_u(k_u) + V_{txa}(k_{txa}) = D(k_d) + E(k_e)$$

Next, subtract $D(k_d)$ from both sides of the equation:

$$V_u(k_u) - D(k_d) + V_{txa}(k_{txa}) = E(k_e)$$

and divide the entire equation by the market value of equity, E:

$$k_e = \frac{V_u}{E}(k_u) - \frac{D}{E}(k_d) + \frac{V_{txa}}{E}(k_{txa})$$

To eliminate V_u from the right side of the equation, rearrange Equation C.1 to $V_u = D - V_{txa} + E$, and use this identity to replace V_u:

$$k_e = \frac{D - V_{txa} + E}{E}(k_u) - \frac{D}{E}(k_d) + \frac{V_{txa}}{E}(k_{txa})$$

Distribute the first fraction into its component parts:

$$k_e = \frac{D}{E}(k_u) - \frac{V_{txa}}{E}(k_u) + k_u - \frac{D}{E}(k_d) + \frac{V_{txa}}{E}(k_{txa}) \tag{C.5}$$

Consolidating terms and rearranging leads to the general equation for the cost of equity:

$$k_e = k_u + \frac{D}{E}(k_u - k_d) - \frac{V_{txa}}{E}(k_u - k_{txa}) \tag{C.6}$$

If debt is a constant proportion of enterprise value (i.e., debt grows as the business grows), k_u will equal k_{txa}. Consequently, the final term drops out:

$$k_e = k_u + \frac{D}{E}(k_u - k_d)$$

We believe this equation best represents the relationship between the levered cost of equity and the unlevered cost of equity.

The same analysis can be repeated under the assumption that the risk of interest tax shields equals the risk of debt. Rather than repeat the first few steps, we start with Equation C.5:

$$k_e = \frac{D}{E}(k_u) - \frac{V_{txa}}{E}(k_u) + k_u - \frac{D}{E}(k_d) + \frac{V_{txa}}{E}(k_{txa})$$

To solve for k_e, replace k_{txa} with k_d:

$$k_e = \frac{D}{E}(k_u) - \frac{V_{txa}}{E}(k_u) + k_u - \frac{D}{E}(k_d) + \frac{V_{txa}}{E}(k_d)$$

Consolidate like terms and reorder:

$$k_e = k_u + \frac{D - V_{txa}}{E}(k_u) - \frac{D - V_{txa}}{E}(k_d)$$

Finally, further simplify the equation by once again combining like terms:

$$k_e = k_u + \frac{D - V_{txa}}{E}(k_u - k_d)$$

The resulting equation is the levered cost of equity for a company whose debt can take any value but whose interest tax shields have the same risk as the company's debt.

Exhibit C.2 summarizes the formulas that can be used to estimate the levered cost of equity. The top row in the exhibit contains formulas that assume k_{txa} equals k_u. The bottom row contains formulas that assume k_{txa} equals k_d. The formulas on the left side are flexible enough to handle any future capital structure but require valuing the tax shields separately. The formulas on the right side assume the dollar level of debt is fixed over time.

EXHIBIT C.2 **Levered Cost of Equity**

	Dollar level of debt fluctuates	Dollar level of debt is constant
Tax shields have same risk as operating assets $k_{txa} = k_u$	$k_e = k_u + \frac{D}{E}(k_u - k_d)$	$k_e = k_u + \frac{D}{E}(k_u - k_d)$
Tax shields have same risk as debt $k_{txa} = k_d$	$k_e = k_u + \frac{D - V_{txa}}{E}(k_u - k_d)$	$k_e = k_u + (1 - T_m)\frac{D}{E}(k_u - k_d)$

Note: k_e = cost of equity
k_d = cost of debt
k_u = unlevered cost of equity
k_{txa} = cost of capital for tax shields
T_m = marginal tax rate
D = debt
E = equity
V_{txa} = present value of tax shields

LEVERED BETA

Similar to the cost of capital, the weighted average beta of a company's assets, both operating and financial, must equal the weighted average beta of its financial claims:

$$\frac{V_u}{V_u + V_{txa}}(\beta_u) + \frac{V_{txa}}{V_u + V_{txa}}(\beta_{txa}) = \frac{D}{D+E}(\beta_d) + \frac{E}{D+E}(\beta_e)$$

Since the form of this equation is identical to the cost of capital, it is possible to rearrange the formula using the same process as previously described. Rather than repeat the analysis, we provide a summary of levered beta in Exhibit C.3. As expected, the first two columns are identical in form to Exhibit C.2, except that the beta (β) replaces the cost of capital (k).

By using beta, it is possible to make one additional simplification. If debt is risk free, the beta of debt is 0, and β_d drops out. This allows us to convert the following general equation (when β_{txa} equals β_u):

$$\beta_e = \beta_u + \frac{D}{E}(\beta_u - \beta_d)$$

into the following:

$$\beta_e = \left(1 + \frac{D}{E}\right)\beta_u$$

EXHIBIT C.3 **Levered Beta**

	Dollar level of debt fluctuates	Dollar level of debt is constant and debt is risky	Debt is risk free
Tax shields have same risk as operating assets $\beta_{txa} = \beta_u$	$\beta_e = \beta_u + \frac{D}{E}(\beta_u - \beta_d)$	$\beta_e = \beta_u + \frac{D}{E}(\beta_u - \beta_d)$	$\beta_e = \left(1 + \frac{D}{E}\right)\beta_u$
Tax shields have same risk as debt $\beta_{txa} = \beta_d$	$\beta_e = \beta_u + \frac{D - V_{txa}}{E}(\beta_u - \beta_d)$	$\beta_e = \beta_u + (1 - T_m)\frac{D}{E}(\beta_u - \beta_d)$	$\beta_e = \left[1 + (1 - T_m)\frac{D}{E}\right]\beta_u$

Note: β_e = beta of equity
β_d = beta of debt
β_u = unlevered beta of equity
β_{txa} = beta of capital for tax shields
T_m = marginal tax rate
D = debt
E = equity
V_{txa} = present value of tax shields

This last equation is an often-applied formula for levering (and unlevering) beta when the risk of interest tax shields (β_{txa}) equals the risk of operating assets (β_u) *and* the company's debt is risk free. For investment-grade companies, debt is nearly risk free, so any errors using this formula will be small. If the company is highly leveraged, however, errors can be large. In this situation, estimate the beta of debt, and use the more general version of the formula.

UNLEVERED BETA AND PENSIONS

Since stockholders are responsible for future pension payments and other retirement obligations, the risks associated with these employee benefits can affect a company's beta. If a company has significant pensions, especially unfunded pensions, make sure to include them in the unlevering process.

If you believe the risk of pension assets matches the risk of future obligations, only the unfunded portion of benefit obligations affects the equity beta. In this case, use the unlevering equations in the preceding sections, but treat any unfunded benefit obligations identically to debt.

If you believe the risk of pension assets does not match the risk of future obligations, the unlevering formulas can be reworked such that the risk of pension assets and risk of benefit obligations are evaluated separately. To do this, start with the portfolio equation for beta:

$$\frac{V_u}{V}\beta_u + \frac{V_{pa}}{V}\beta_{pa} = \frac{V_{pbo}}{V}\beta_{pbo} + \frac{D}{V}\beta_d + \frac{E}{V}\beta_e$$

where V_{pa} = value of pension assets

V_{pbo} = present value of pension benefit obligations

β_{pa} = beta of pension assets

β_{pbo} = beta of pension benefit obligations

V = sum of debt, equity, and benefit obligations

Next, multiply both sides by V:

$$V_u\,\beta_u + V_{pa}\,\beta_{pa} = V_{pbo}\,\beta_{pbo} + D\beta_d + E\beta_e$$

Subtract the term related to pension assets from both sides of the equation:

$$V_u\,\beta_u = V_{pbo}\,\beta_{pbo} + D\beta_d + E\beta_e - V_{pa}\,\beta_{pa}$$

To isolate β_u, divide both sides by V_u. This leads to the general equation for estimating unlevered beta, inclusive of pensions:

$$\beta_u = \frac{V_{pbo}}{V_u}\beta_{pbo} + \frac{V_d}{V_u}\beta_d + \frac{V_e}{V_u}\beta_e - \frac{V_{pa}}{V_u}\beta_{pa}$$

If debt and pension liabilities have the same beta, simplify the last equation by combining terms:

$$\beta_u = \frac{D + V_{pbo}}{V_u}\beta_d + \frac{E}{V_u}\beta_e - \frac{V_{pa}}{V_u}\beta_{pa}$$

Chapter 23 discusses how to incorporate pensions into a valuation. In Exhibit 23.5, we use the equation above to unlever beta for a set of food manufacturers. Given the small size of each company's pension relative to the respective company's market value of equity, the difference in unlevered beta with and without the pension adjustment is minor. We therefore recommend adjusting beta for pensions only when pension assets and liabilities are substantial. In most situations, the unlevering equations that classify the unfunded portion of pensions as debt will suffice.